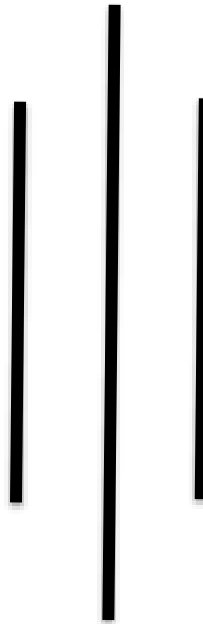




Tribhuvan University

Institute of Science and Technology

Amrit Campus



A Lab Report on
Design and Analysis of Algorithm (CSC-325)

Submitted to:
Department of Computer Science & Information Technology
Amrit Campus, Thamel, Kathmandu

Tribhuvan University

In partial fulfillment of the requirement for the *Bachelor's Degree in*
Computer Science & Information Technology

Submitted By:

Name: Krishna Aryal
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Section: B

Submitted To:

Asst. Prof. Dabbal Singh Mahara
Department of Computer Science and
Information Technology

Submission Date:

Signature

LABWORKS

S.No.	Name of Experiments	Date of Submission	Signature	Remarks
1.	<u>Implementation of Iterative Algorithms</u> Task1: Generating Fibonacci Series. Task2: Calculating GCD using Euclidean Algorithm. Task3: Implementing Sequential Search. Task4: Implementing Bubble Sort. Task5: Implementing Selection Sort Task6: Implementing Insertion Sort.			
2.	<u>Implementing Divide and Conquer Algorithms</u> Task1: Binary Search Task2: Min-Max Finding Task3: Merge Sort Task4: Quick Sort Task5: Heap Sort			
3.	<u>Implementing Greedy Algorithms</u> Task1: Implementing Fractional Knapsack Task2: Implementing Kruskal's Algorithm to find MST Task3: Implementing Dijkstra's Algorithm Task4: Implementing Huffman Coding Algorithm			
4.	<u>Implementing Dynamic Programming Algorithms</u> Task1: Implementing Matrix Chain Multiplication to determine minimum number of scalar multiplications required. Task2: Implementing 0/1 Knapsack Problem to determine maximum profit. Task3: Implementing Floyd-Warshall Algorithm to find shortest paths between all pairs of vertices.			
5.	<u>Implementing Backtracking Algorithms</u> Task1: Implementing Subset Sum Problem. Task2: Implementing 0/1 Knapsack Problem. Task3: Implementing N-Queen Problem.			
6.	<u>Implementing Number Theoretic Algorithms</u> Task1: Implementing Chinese Remainder Theorem. Task2: Implementing Miller-Rabin Primality Test.			
7.	<u>Implementing Approximation Algorithms</u> Task1: Implementing Vertex Cover Problem. Task2: Implementing Subset Sum Problem.			